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INFORMATION PROCESSING APPARATUS, METHOD FOR CONTROLLING INFORMATION PROCESSING APPARATUS, AND STORAGE MEDIUM

Abstract

An information processing apparatus comprising at least one memory and at least one processor which function as a consolidation unit that configures a setting for consolidating a plurality of sets of image data on a single sheet and a determination unit that automatically determines a position to perforate the sheet based on the setting configured by the consolidation unit.

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Background/Summary

BACKGROUND

Field

[0001] The present disclosure relates to an information processing apparatus, a method for controlling the information processing apparatus, and a storage medium.

Description of the Related Art

[0002] Conventionally, some generally used sheets can be perforated to be tearable. Some multifunctional devices are known to have the function of setting perforation positions on a sheet and perforating the sheet in execution of copy or print.

[0003] Japanese Patent Laid-Open No. 2019-81318 discloses a technique of displaying perforation lines on a preview of a scanned original and allowing the positions of the perforation lines to be adjusted.

SUMMARY

[0004] What is in demand is a technique for allowing a user to configure sheet processing settings more easily on a setting screen.

[0005] An information processing apparatus according to an aspect of the present disclosure has a consolidation unit that configures a setting for consolidating a plurality of sets of image data on a single print sheet and a determination unit that automatically determines a position to perforate the sheet based on the setting configured by the consolidation unit.

[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a system configuration diagram;

[0008] FIG. 2 is a hardware configuration diagram of an image processing apparatus;

[0009] FIG. 3 is a software configuration diagram of the image processing apparatus;

[0010] FIG. 4 is a diagram showing a screen showing a list of setting items for copy or print;

[0011] FIG. 5 is a diagram showing a page consolidation setting screen;

[0012] FIG. 6 is a flowchart of perforation setting;

[0013] FIG. 7 is a flowchart of automatic perforation position determination;

[0014] FIG. 8 is a flowchart of automatic perforation position determination;

[0015] FIG. 9 is a diagram showing an image repeat setting screen;

[0016] FIG. 10 is a flowchart of automatic perforation position determination;

[0017] FIG. 11 is a diagram showing a setting screen for manually specifying perforation positions;

[0018] FIG. 12 is a diagram showing a screen displaying a list of specified perforation positions;

[0019] FIG. 13 is a flowchart of control performed in a case where an earlier operation is prioritized;

[0020] FIG. 14 is a flowchart of control performed in a case where an earlier operation is prioritized;

[0021] FIG. 15 is a flowchart of control performed in a case where a later operation is prioritized;

[0022] FIG. 16 is a flowchart of control performed in a case where a later operation is prioritized;

[0023] FIG. 17 is a diagram showing a setting screen for switching a setting regarding specification of perforation positions;

[0024] FIG. 18 is a diagram showing a software configuration diagram of an information processing apparatus;

[0025] FIG. 19 is a diagram showing a printer driver's setting screen; and
[0026] FIGS. 20A and 20B are diagrams illustrating a concrete example of an image consolidation function.

DESCRIPTION OF THE EMBODIMENTS

[0027] The present disclosure is described in detail below with reference to the drawings. Note that embodiments below are not to limit the present disclosure according to the scope of claims, and not all the combinations of the features described in the embodiments are necessarily essential as solutions provided by the present disclosure. Also, throughout the drawings attached hereto, the same or like configurations are denoted by the same reference number to omit repetitive descriptions.

Embodiment 1

<System Configuration>

[0028] FIG. 1 is a schematic diagram showing the configuration of a printing and processing system having an image processing apparatus according to the present embodiment. The printing and processing system of the present embodiment is a system formed by an image processing apparatus **103**, a server personal computer (PC) **101** being an information processing apparatus, and a client PC **102** being an information processing apparatus, which are connected via a network.

[0029] The image processing apparatus **103** is a multifunction peripheral (MFP). The image processing apparatus **103** has the function of copying a paper original and the function of printing print data sent from an external printer driver. The image processing apparatus **103** also has the function of scanning a paper original and sending scanned image data to an external file server (the server PC **101**) or to an email address (the Send function or the Scan-to-Send function). The image processing apparatus **103** also has the function of transmitting data to another image processing apparatus to have the data printed by the destination image processing apparatus (the Remote Copy function or the Fax function). The image processing apparatus **103** may have other various functions. Note that the functions described above are merely examples, and the functions of the image processing apparatus **103** are not limited to them. The image processing apparatus **103** may be an apparatus having a single function. The image processing apparatus **103** used in the present embodiment may employ the electrophotographic printing method, the inkjet printing method, or other printing methods.

[0030] The server PC **101** or the client PC **102** can configure settings related to printing or processing using a printer driver and instruct the image processing apparatus **103** to print or process via a network set.

[0031] The image processing apparatus **103** is connected to the server PC **101** and the client PC **102** by, for example, Ethernet (registered trademark) herein, but the present disclosure is not limited to this example.

[0032] Also, although a plurality of apparatuses are described as an example, the server PC **101** and the client PC **102**, for example, may be formed by the same computer. It is also possible to incorporate the server PC **101** and the client PC **102** into the image processing apparatus **103** so that the system **100** is formed by the image processing apparatus **103** alone. Although the image processing apparatus **103** is shown as an example of the information processing apparatus in the present embodiment, it is to be noted that the information processing apparatus is not limited to the image processing apparatus **103** and may be a PC, a terminal apparatus, or a smartphone.

<Hardware Configuration of the Image Processing Apparatus>

[0033] FIG. 2 is a block diagram showing a hardware configuration of the image processing apparatus **103** according to the present embodiment. A controller unit **200** is connected to a scanner **270**, which is an image input device, or a printer **295**, which is an image output device. Meanwhile, the controller unit **200** is connected to Ethernet (registered trademark) or a public network to receive and output image information and device information.

[0034] A CPU **201** is a controller that controls the image processing apparatus **103**. A RAM **202** is

system work memory used for the CPU **201** to operate and is also image memory to temporarily store image data. A ROM **203** is a boot ROM to store a boot program for the system. An HDD **204** is a hard disk drive where system software, applications, and image data are stored.

[0035] An operation unit I/F **206** is an interface with an operation unit **212** having a touch panel and outputs image data to be displayed on the operation unit **212** to the operation unit **212**. The operation unit I/F **206** also serves the role of conveying information entered on the operation unit **212** by a system user to the CPU **201**. A network I/F **210** is connected to the network and enables input and output of information. A modem **250** is connected to the public network and enables input and output of information.

[0036] An SRAM **209** is a non-volatile storage medium capable of high speed operation. An RTC **211** is a real-time clock and keeps counting the current time even with no power being supplied to the controller unit **200**. The aforementioned devices are disposed on a system bus **207**.

[0037] An image bus I/F **205** is a bus bridge connecting the system bus **207** and an image bus **208**, which transfers image data at high speed, to convert data structures. The image bus **208** is configured by a PCI bus or IEEE 1394. The following devices are disposed on the image bus **208**.

[0038] A RIP **260** is a raster image processor and converts PDL code into an bitmap image. A device I/F unit **220** is connected to the scanner **270** and the printer **295**, which are image input and output devices, and to the controller unit **200** and performs conversion of image data between synchronous and asynchronous systems. A scanner image processing unit **280** corrects, processes, and edits input image data. A printer image processing unit **290** performs printer corrections, resolution conversion, and the like on print output image data. An image rotation unit **230** rotates image data. An image compression/decompression unit **240** performs compression processing and decompression processing.

[0039] A perforation unit **296** is connected to the controller unit **200** via the device I/F unit **220** and processes a sheet outputted from the printer image processing unit **290** by making dotted lines of perforation in the sheet. Although the image processing apparatus **103** of the present embodiment includes the perforation unit **296** so that a sheet can be perforated in the image processing apparatus **103**, it is to be noted that printing and processing may be executed by different apparatuses. In this case, for example, a sheet printed by the image processing apparatus **103** is conveyed to another apparatus where the sheet is perforated.

<Software Configuration of the Image Processing Apparatus>

[0040] FIG. **3** is a block diagram showing a software configuration of the image processing apparatus **103** according to the invention of the present application. This software is stored in the HDD **204** of the image processing apparatus **103** as firmware **350** and is loaded into the RAM **202** and executed by the CPU **201**.

[0041] A real-time OS **301** is a real-time operating system and provides software running thereon with a service and its framework for management of various resources optimized for control of an embedded system. The service and framework for management of various resources provided by the real-time OS includes: multitask (thread) management that substantially enables a plurality of processes to operate in tandem by managing a plurality of processing execution contexts by the CPU; inter-task communications that enables synchronization or data exchange between tasks; and a protocol stack implementing memory management, interrupt management, various device drivers, or various protocols' processing (such as a local interface, a network, or communication).

[0042] A controller platform **302** is formed of a file system **303**, a job/device control **304**, a counter **305**, and the like. The file system **303** is a mechanism for storing data built on a storage device such as the HDD **204** or the RAM **202**. The job/device control **304** controls the hardware of the image processing apparatus **103** and also controls jobs that use the basic functions (such as print, scan, communication, and image conversion) provided mainly by the hardware of the image processing apparatus **103**. The counter **305** manages counter values related to expiry of each application, printing, or scanning, which are stored in the SRAM **209**.

[0043] A system service **306** is formed by application management **308**, window management **309**, and the like. The application management **308** is a management module for installing, uninstalling, starting, or stopping, e.g., applications **310** to **314**. The applications are, for example, home **310**, print **311**, copy **312**, scan **313**, and login **314**. The window management **309** is a management module for controlling the order of displaying application windows of the applications.

[0044] An application platform **307** is middleware for allowing the real-time OS **301** and the mechanisms of the controller platform **302** to be used from each application.

[0045] The applications are applied software implementing various business models by using functions provided by the application platform **307**. The home **310** includes a menu function for selecting an application. The print **311** is a print function for printing a received print job of digital data on a sheet. The copy **312** is a copy function for making a copy of a paper original. The scan **313** is a scan function for sending data. Upon receipt of a login notification, each application performs screen generation processing appropriate for the logged-in user based on information on the notification. For example, in an example where a logged-in user is prohibited from using the web browser or from making a color copy, the home **310** performs processing so that the logged-in user cannot select the web browser that they are not allowed to use or performs processing so that the logged-in user cannot select a button for color copy.

[0046] The login **314** is authentication software for performing user authentication. The login **314** performs login processing based on a username and a password entered on the operation unit **212**. In a case where the authentication fails, an error message is displayed, and in a case where the authentication succeeds, a series of steps to be performed after authentication is performed.

[0047] The image processing apparatus **103** has an image consolidation function for printing a plurality of images on a single sheet. This image consolidation function includes, for example, what is called the page consolidation function or the N-up function, where N originals are allocated and printed onto a single sheet. The image consolidation function also includes what is called an image repeat function, where a plurality of images are allocated and printed onto a single sheet according to settings to print a particular image X times vertically and Y times horizontally. Note that X and Y above are natural numbers, and thus, the image repeat function of the present embodiment may be configured to print the same image repeatedly only vertically or horizontally. These image consolidation functions may be used for usage where, for example, images on a sheet may be torn off piece by piece and distributed, such as coupons. Thus, in a case where the image consolidation function is set, it is convenient for a user to be able to select, on the same setting screen, the perforation setting for perforating the sheet for every image, and this may offer improved efficiency in configuring settings.

[0048] The exiting methods require the image consolidation function and the perforation setting to be set on different screens, and improvement in efficiency is called for from the perspective of efficiency in configuring settings. Also, in an existing known perforation setting method, the position to draw a perforation line on a sheet is specified manually with a numerical value. Also, to specify a plurality of perforation lines, the above operation needs to be repeated for each perforation line, which is problematic in terms of efficiency in configuring settings. Also, in order to accurately specify the position of a perforation line manually with a numerical value, it is necessary to measure the sheet with a measuring device such as a scale and check the position to draw a perforation line in advance before printing, which is troublesome.

[0049] The present embodiment described below proposes a method where for each image, the setting of the image consolidation function and the setting of sheet perforation can be set on the same setting screen. In the present embodiment, perforation positions are automatically determined based on the image consolidation function. Thus, even in a case where there are a plurality of perforation lines to specify, less trouble is involved in configuring settings. Specifically, the present embodiment eliminates the trouble of measuring a sheet with a scale or the like beforehand and determining perforation positions accurately.

<Screens Displayed According to Embodiment 1>

[0050] Next, a control method of Embodiment 1 is described.

[0051] FIG. 4 is a diagram showing a screen with a list of setting items of the present embodiment. A list screen **401** shown in FIG. 4 shows some of the functions which may optionally be set by a user for execution of the copy **312**. Note that the list screen **401** shown in FIG. 4 can be applied to the print **311** as well. The setting items list screen **401** of the present embodiment is outputted from the window management **309** to the operation unit **212**. The window management **309** is stored as the firmware **350** in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. The setting items list screen **401** has setting items to be configured for copy or print, such as, for example, page consolidation **402**, image repeat **403**, and perforation position specification **404**.

[0052] For example, upon detection of selection of the page consolidation **402**, the CPU **201** performs display control to display a page consolidation function setting screen **501** shown in FIG. 5 on the operation unit **212**. Also, upon detection of selection of, for example, the image repeat **403**, the CPU **201** displays an image repeat function setting screen **901** shown in FIG. 9 to be described later.

[0053] FIG. 5 is a diagram showing a setting screen for the image consolidation function. Specifically, FIG. 5 shows the page consolidation function setting screen **501** for setting the function of allocating and printing a plurality of originals onto a single sheet, which is called N-up. The page consolidation function setting screen **501** of the present embodiment is outputted from the window management **309** to the operation unit **212**.

[0054] The window management **309** is stored as the firmware **350** in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. The page consolidation function setting screen **501** has setting items to be set for execution of the print **311** or the copy **312**, and N (N is specified as a multiple of 2) images are allocated and printed on a single sheet based on the settings configured. The page consolidation function setting screen **501** has N-up set values **502** for setting how many images to allocate to a single sheet.

[0055] For example, an 8-up setting value **503** is one of the N-up set values **502**, and in a case where the 8-up setting value **503** is set, eight original images are printed on a single sheet. The page consolidation function setting screen **501** has, separately from the N-up set values **502** for page consolidation, a checkbox **504** for automatically setting perforation positions, which is used to set perforation lines for the sheet so that each image allocated to the sheet can be torn off. In a case where any of the N-up set values **502** is enabled, the checkbox **504** for automatically setting perforation positions can be used to enable or disable (i.e., switch) the setting for perforating the sheet in between the allocated pages. A checkbox **505** for setting margins is an entry field which, by being enabled, can add margins of preset lengths to a print sheet. Once an OK button **506** disposed on the page consolidation function setting screen **501** is pressed, set values, such as the N-up set value **502** and the checkbox **504** for automatically setting perforation positions, are recorded in the RAM **202** or the HDD **204**.

<Flowcharts According to Embodiment 1>

[0056] FIG. 6 is a flowchart for performing copy or print according to the present embodiment. Specifically, in the flowchart, it is determined whether the perforation setting is set in addition to the image consolidation function setting for printing N images on a single sheet, and processing is executed in accordance with the determination. The firmware **350** according to the processing of the present embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. The present processing starts upon detection of an operation for starting copy or print. Also, the letter “S” in the description of each process in the flowchart means that it is a step in the flowchart, and this applies to the rest of the flowcharts herein.

[0057] Although using the page consolidation function is used as one of the image consolidation

functions in the example described in the present flowchart, it is to be noted that the flowchart can in fact be applied to processing for using image repeat, which will be described in a later embodiment.

[0058] In **S601**, the CPU **201** starts copy or print in response to detection of copy or print execution processing. In **S602**, the CPU **201** determines whether the image consolidation function of printing a plurality of images on a single sheet is set. Specifically, the CPU **201** determines whether either the page consolidation **402** or the image repeat **403** is selected and set on the list screen **401** shown in FIG. **4**. If it is determined that the image consolidation function is not set, the CPU **201** proceeds to **S603** to execute copy or print. After that, the CPU **201** ends the processing of the present flowchart. By contrast, if it is determined that the image consolidation function is set, the CPU **201** proceeds to **S604** to determine whether the perforation setting is set on the setting screen for the image consolidation function selected. If it is determined that the perforation setting is not set, the CPU **201** proceeds to **S605** to perform copy or print with the image consolidation function being applied. By contrast, if it is determined that the perforation setting is set, the CPU **201** proceeds to **S606**.

[0059] In **S606**, the CPU **201** determines perforation positions. Details of the perforation position determination in **S606** will be described later using FIG. **7**. In **S607**, the CPU **201** performs copy or print according to the settings of the image consolidation function and the perforation setting determined in **S602** and **S606**. After that, the CPU **201** ends the processing of the present flowchart.

[0060] FIG. **7** is a flowchart of the perforation position determination according to the present embodiment, or specifically, a flowchart showing details of **S606** in FIG. **6** described above. The firmware **350** according to the processing of the present embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**.

[0061] Note that the present processing is a broad concept for both of processing for the page consolidation function, which has been described thus far in the present embodiment, and processing for image repeat, which will be described in a later embodiment. In the present embodiment, first, FIG. **7** is used to describe an overview of processing for adding perforation, and next, FIG. **8** is used to describe a flowchart of processing performed in a case where the processing in FIG. **7** is applied to the page consolidation function (the N-up function). Also, a flowchart of processing performed in a case where the processing in FIG. **7** is applied to the image repeat function will be described later using FIG. **10**.

[0062] In **S701**, the CPU **201** obtains the vertical and horizontal sizes of a sheet used for print from the RAM **202**. The present processing is also called sheet size obtainment processing.

[0063] In **S702**, the CPU **201** obtains an image repeat count set for the image consolidation function from the RAM **202**. The present processing is also called repeat count obtainment processing.

[0064] In **S703**, the CPU **201** determines the position of each horizontal perforation line based on the vertical size of the sheet obtained in the sheet size obtainment processing in **S701** and the image repeat count obtained in the repeat count obtainment processing in **S702**. The present processing is also called horizontal perforation setting processing.

[0065] In **S704**, the CPU **201** determines the position of each vertical perforation line based on the horizontal size of the sheet obtained in the sheet size obtainment processing in **S701** and the image repeat count obtained in the repeat count obtainment processing in **S702**. The present processing is also called vertical perforation setting processing. After that, the CPU **201** ends the processing of the present flowchart. Note that the sheet size obtainment processing in **S701** and the repeat count obtainment processing in **S702** may be reversed in order. Also, the horizontal perforation setting processing in **S703** and the vertical perforation setting processing in **S704** may be reversed in order.

[0066] FIG. **8** is a flowchart of perforation position determination according to the present embodiment, or specifically, a flowchart showing details of **S606** in FIG. **6** described earlier. Also, FIG. **8** is a flowchart performed in a case where the perforation position determination processing

described using FIG. 7 is applied to the N-up function. The firmware 350 according to the processing of the embodiment is stored in the HDD 204 of the image processing apparatus 103 and is loaded into the RAM 202 and executed by the CPU 201.

[0067] In S801, the CPU 201 obtains the vertical and horizontal sizes of a sheet used for print from the RAM 202.

[0068] In S802, the CPU 201 obtains a division number (the value of N in N-up) from the RAM 202. Note that N is a multiple of 2.

[0069] In S803, the CPU 201 sets a horizontal perforation line to the sheet from a position corresponding to a length obtained by dividing the vertical size of the sheet obtained in S801 by $N/2$ using the value of N obtained in S802, and repeats this ($N/2-1$) times. Then, the CPU 201 determines those positions as the positions of horizontal perforation lines. Note that in a case where margins are set for the sheet to be printed, the CPU 201 may set a horizontal perforation line to the sheet from a position corresponding to a length obtained by dividing the vertical size of the sheet minus the margins by $N/2$, repeat this ($N/2-1$) times, and determine those positions as the positions of horizontal perforation lines.

[0070] In S804, the CPU 201 determines the position of a vertical perforation line by setting a vertical perforation line to the sheet at a position corresponding to a length obtained by dividing the horizontal size of the sheet obtained in S801 by 2. After that, the CPU 201 ends the processing of the present flowchart. Note that in a case where margins are set for the sheet to be printed, the CPU 201 may determine the position of a vertical perforation line by setting a vertical perforation line to the sheet from a position corresponding to a length obtained by dividing the horizontal size of the sheet minus the margins by 2.

[0071] FIGS. 20A and 20B are diagrams illustrating concrete examples of the image consolidation function. A concrete example of the flowchart of the N-up perforation position determination described above is described with reference to FIG. 20A.

[0072] It is assumed here that the sheet size obtained by the CPU 201 in S801 is “8” as the vertical size and “4” as the horizontal size. It is also assumed here that the division number N obtained by the CPU 201 in S802 is “8.” This setting means that eight originals are allocated to a sheet with a size of 8×4 .

[0073] In S803, the CPU 201 determines the positions of horizontal perforation lines. Specifically, the CPU 201 sets a horizontal (X-direction) perforation line from the position of “2,” which is a length obtained by dividing the vertical sheet size “8” by “8/2.” The CPU 201 repeats this ($8/2-1$) times (i.e., three times). With the coordinates of the lower left corner of the sheet being (0, 0), perforation lines are set from the positions (0, 2), (0, 4), and (0, 6) to the positions (4, 2), (4, 4), and (4, 6), respectively.

[0074] In S804, the CPU 201 determines the position of a vertical perforation line. Specifically, the CPU 201 sets a vertical (Y-direction) perforation line from a position (2, 0) corresponding to a length obtained by dividing the horizontal sheet size “4” by “2.” With the coordinates of the lower left corner of the sheet being (0, 0), a perforation line is set from the position (2, 0) to the position (2, 8).

[0075] As thus described, for 8-up, a single vertical perforation line and three horizontal perforation lines are drawn, and the sheet is divided into eight equal parts by the perforation lines.

[0076] As an alternative way, values of perforation positions determined beforehand for various sheet sizes and various N-up settings may be stored in the HDD 204 or the RAM 202 in a table format to replace S606. For instance, in a case where 4-up is set for an A4 standard-size sheet measuring 297 mm vertically and 210 mm horizontally, values stored in a table may indicate that a vertical perforation line is drawn at the position 148.5 mm and a horizontal perforation line is drawn at the position 105 mm. The processing in S607 may be executed in reference to the values of the perforation positions in the table stored in the HDD 204 or the RAM 202.

[0077] As thus described, the preset embodiment can improve user convenience regarding their

experience on the setting screen. Specifically, the user can set the perforation setting on the setting screen for the image consolidation function. In a case where a user sets the perforation setting, the positions of perforation lines are automatically calculated based on the sheet size and the N-up division setting. This is easier and more convenient and also more efficient than a conventional method where perforation positions are manually specified with numerical values. Also, compared to the method where perforation positions are manually specified with numerical values, the present embodiment saves the trouble of measuring the size of a sheet beforehand with a scale or the like and the trouble of entering accurate numerical values according to those measurement values.

Embodiment 2

[0078] Embodiment 2 has the same basic configuration as Embodiment 1.

<Screen Displayed According to Embodiment 2>

[0079] FIG. 9 is a diagram showing an image consolidation function setting screen according to the present embodiment. Specifically, FIG. 9 shows the image repeat function setting screen **901** for setting what is called the image repeat function, where a single image is printed repeatedly X times vertically and Y times horizontally on a single sheet. The image repeat function setting screen **901** of the present embodiment is outputted from the window management **309** to the operation unit **212**. The window management **309** is stored as the firmware **350** in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**.

[0080] The image repeat function setting screen **901** has setting items to be set in execution of the print **311** or the copy **312**, and a particular image is printed on a single sheet repeatedly X times vertically and Y times horizontally based on the settings configured. The image repeat function setting screen **901** has image repeat count set values **902** for setting how many times to allocate (repeat) the same image on a single sheet vertically and horizontally. For example, a vertical image set value **903** is one of the set values **902**, and the same image is printed on a single sheet repeatedly X times vertically based on the numerical value X specified in the vertical image set value **903** using increase and decrease buttons. Also, in a case where a vertical automatic setting **904** is set, the value specified in the vertical image set value **903** is ignored, and the maximum number of images repeatable on a sheet is automatically calculated.

[0081] The image repeat function setting screen **901** has, separately from the image repeat set values **902**, a checkbox **905** for automatically setting perforation positions, which is an entry field for setting perforation lines along which a sheet can be torn to separate the images allocated to the sheet one by one. In a case where the image repeat set values **902** are enabled, the checkbox **905** for automatically setting perforation positions can enable or disable (i.e., switch) the setting for perforating the sheet in between allocated images. A checkbox **906** for setting margins is an entry field which, by being enabled, can set margins of preset lengths to a print sheet. Once an OK button **907** disposed on the image repeat function setting screen **901** is pressed, set values such as the image repeat count set values **902** and the checkbox **905** for automatically setting perforation positions are recorded in the RAM **202** or the HDD **204**.

<Flowchart According to Embodiment 2>

[0082] FIG. 10 is a flowchart of perforation position determination according to the present embodiment, or specifically, a flowchart showing the details of S606 in FIG. 6 described earlier. The flowchart corresponds to the perforation position determination processing described using FIG. 7 and is executed in a case where it is determined that the perforation setting is set in addition to the image repeat function as one of image consolidation functions of printing N images on a single sheet. The firmware **350** according to the processing of the embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**.

[0083] In S1001, the CPU **201** obtains the vertical and horizontal sizes of a sheet used for print from the RAM **202**. In S1002, the CPU **201** obtains, from the RAM **202**, an image repeat count X

for repeating the image vertically and an image repeat count Y for repeating the image horizontally (X and Y are natural numbers).

[0084] In **S1003**, the CPU **201** determines the position of a horizontal perforation. Specifically, the CPU **201** sets a horizontal perforation line from a position corresponding to a length obtained by dividing the vertical sheet size obtained in **S1001** by X, which is the vertical image repeat count obtained in **S1002**, repeats this (X-1) times, and determines those positions as the positions of horizontal perforation lines.

[0085] In **S1004**, the CPU **201** determines the positions of vertical perforation lines. Specifically, the CPU **201** sets a vertical perforation line from a position corresponding to a length obtained by dividing the horizontal sheet size obtained in **S1001** by Y, which is the horizontal image repeat count obtained in **S1002**, repeats this (Y-1) times, and determines those positions as the positions of vertical perforations lines. After that, the CPU **201** ends the processing of the present flowchart.

[0086] With reference to FIG. **20B**, a concrete example of a perforation position determination flowchart for the image repeat function described above is described.

[0087] It is assumed here that the sheet size obtained by the CPU **201** in **S801** are “8” as the vertical size and “4” as the horizontal size. It is also assumed here that in **S1002**, the CPU **201** obtains “8” as a vertical image repeat count and “4” as a horizontal repeat count from the RAM **202**.

[0088] In **S1003**, the CPU **201** determines the positions of horizontal perforation lines. Specifically, the CPU **201** sets a horizontal (X-direction) perforation line from a position corresponding to a length obtained by dividing the vertical sheet size “8” by the vertical image repeat count “8” and repeats this (8-1) times (i.e., seven times). Specifically, with the coordinates of the lower left corner of the sheet being (0, 0), perforation lines are set from the positions (0, 1) to (0, 7) to the positions (4, 1) to (4, 7), respectively.

[0089] In **S1004**, the CPU **201** determines the positions of vertical perforation lines. Specifically, the CPU **201** sets a vertical (Y-direction) perforation line from a position corresponding to a length obtained by dividing the horizontal sheet size “4” by the horizontal image repeat count “4” and repeats this (4-1) times (i.e., three times). With the coordinates of the lower left corner of the sheet being (0, 0), perforation lines are set from the positions (1, 0), (2, 0), and (3, 0) to the positions (1, 8), (2, 8), and (3, 8), respectively.

[0090] As thus described, for the image repeat of the above settings, three vertical perforation lines and seven horizontal perforation lines are set, and the sheet is divided into 32 equal parts by the perforation lines.

[0091] As thus described above, in a case of using the image repeat function as well, the positions of perforation lines can be automatically set like in Embodiment 1. In a case where a plurality of images are allocated to a single sheet, using a suitable perforation calculation method offers expandability for the printing.

Embodiment 3

[0092] Next, Embodiment 3 is described. Embodiment 3 has the same basic configuration as Embodiments 1 and 2 described above.

<Screens Displayed According to Embodiment 3>

[0093] FIG. **11** is a manual perforation position setting screen **1101** showing a method of manually specifying perforation positions according to the present embodiment. The manual perforation position setting screen **1101** is a setting screen to which the list screen **401** in FIG. **4** transitions upon selection of the perforation position specification **404**. The manual perforation position setting screen **1101** of the present embodiment is outputted from the window management **309** to the operation unit **212**. The window management **309** is stored as the firmware **350** in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. The manual perforation position setting screen **1101** has setting items to be set in execution of the print **311** or the copy **312**, and perforation lines are drawn at particular positions on a single

sheet based on the settings configured. The manual perforation position setting screen **1101** has a setting item **1102** for determining whether to draw a vertical perforation line or to draw a horizontal perforation line. In a case where vertical is selected, a perforation line is drawn vertically relative to the sheet, and in a case where horizontal is selected, a perforation line is drawn horizontally relative to the sheet. The manual perforation position setting screen **1101** has a setting item **1103** for specifying the position to draw a perforation line. A position to draw a perforation line is specified using numerical value increase and decrease buttons **1104** and is reflected in the setting item **1103**. The set value is recorded in the HDD **204** upon pressing of a Determine Positions button **1105** disposed on the manual perforation position setting screen **1101**.

[0094] FIG. **12** is a perforation settings list screen **1201** showing manually specified perforation positions according to the present embodiment. The perforation settings list screen **1201** of the present embodiment is outputted from the window management **309** to the operation unit **212**. The window management **309** is stored as the firmware **350** in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. The perforation settings list screen **1201** has a list **1202** which lists perforation lines that are already set. A single perforation line already set is displayed on the list as a single perforation line item **1203**.

[0095] Once any of the perforation line items **1203** is pressed, the perforation settings list screen **1201** transitions to the manual perforation position setting screen **1101** shown in FIG. **11** to display details of the perforation line item, including numerical values representing the position of the already-set perforation line. Specifically, in execution of the print **311** or the copy **312**, upon detection of selection of the perforation position specification **404** in the function list shown in FIG. **4**, the CPU **201** changes the screen to the perforation settings list screen **1201** shown in FIG. **12**. Then, once any of the perforation line items **1203** is pressed on the perforation settings list screen **1201**, the screen transitions to the manual perforation position setting screen **1101** shown in FIG. **11**. In a case where an Add New Perforation Line button **1204** in the list **1202** is pressed, the CPU **201** displays default settings, i.e., blank settings.

[0096] Note that in a case where set values for adding perforation in between allocated pages have been set previously, the list screen **1201** may display, in the list **1202**, all the perforation positions determined in **S606**. More specifically, for example, in a case where the checkbox **504** for automatically setting perforation positions is previously checked on the page consolidation function setting screen **501** shown in FIG. **5** for the allocated pages, the set perforation positions may be displayed in the list **1202**. Also, upon pressing of any of the perforation line items **1203** displayed in the list **1202**, the screen may transition to the manual perforation position setting screen **1101** as described above, and perforation position settings already automatically set can be manually changed there using numerical values. Set values are recorded in the HDD **204** upon pressing of an OK button **1205** disposed on the perforation settings list screen **1201**.

<Flowcharts According to Embodiment 3>

[0097] FIGS. **13** and **14** show control performed in a case where conflicting settings are set and a setting configured earlier is prioritized.

[0098] FIG. **13** is a flowchart of screen control according to the present embodiment performed in a case where a user wants to manually set perforation settings even though the perforation setting is set in addition to the setting for printing a plurality of images on a single sheet. The firmware **350** according to the processing of the present embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. Note that the present processing starts upon detection of an operation for configuring settings for execution of the print **311** or the copy **312**.

[0099] In **S1301**, the CPU **201** detects an attempt to display a setting screen for the image consolidation function (the page consolidation **402** or the image repeat **403** in FIG. **4**). This may be detecting, for example, pressing of the page consolidation **402** or the image repeat **403** in FIG. **4**.

[0100] In **S1302**, the CPU **201** determines whether perforation settings have already been manually

configured on a different screen. The different screen is, for example, the manual perforation position setting screen **1101** or the perforation settings list screen **1201**.

[0101] If it is determined that perforation settings have not already been manually configured, the CPU **201** proceeds to **S1303** to display the setting screen for the image consolidation function that the user tried to display in **S1301**, with an entry field for the perforation setting being not grayed out. The entry field is the checkbox **504** for automatically setting perforation positions in a case of the page consolidation setting screen in FIG. 5 and is the checkbox **905** for automatically setting perforation positions in a case of the image repeat function setting screen **901** in FIG. 9.

[0102] By contrast, if it is determined that perforation settings have already been manually configured, the CPU **201** proceeds to **S1304** to display the setting screen for the image consolidation function that the user tried to display in **S1301**, with an entry field for the perforation setting being grayed out. Specifically, for example, in a case of the page consolidation setting screen in FIG. 5, the CPU **201** displays the setting screen with the checkbox **504** for automatically setting perforation positions being grayed out so as not to accept an operation for switching the perforation setting between enabled and disabled.

[0103] More specifically, in a case where the page consolidation function setting screen **501** shown in FIG. 5 is displayed with perforation settings having already been manually configured, the CPU **201** displays the setting screen with the checkbox **504** for automatically setting perforation positions being grayed out. Also, in a case where the image repeat function setting screen **901** shown in FIG. 9 is displayed with perforation settings having already been manually configured, the CPU **201** displays the image repeat function setting screen **901** with the checkbox **905** for automatically setting perforation positions being grayed out.

[0104] FIG. 14 is a flowchart of screen control according to the present embodiment performed in the following case: with perforation settings being previously manually configured, a user later tries to manually configure perforation settings in addition to the setting for printing a plurality of images on a single sheet. The firmware **350** according to the processing of the embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. Note that the present processing starts upon detection of an operation for configuring settings for execution of the print **311** or the copy **312**.

[0105] In **S1401**, the CPU **201** detects an attempt to display a screen for manually configuring perforation settings. The screen may be, for example, the manual perforation position setting screen **1101** or the perforation settings list screen **1201**.

[0106] In **S1402**, the CPU **201** determines whether the automatic perforation setting has already been set on a different screen in addition to the setting for printing a plurality of images on a single sheet. Specifically, for example, the CPU **201** determines whether the page consolidation function is set on the page consolidation function setting screen **501** with the checkbox **504** for automatically setting perforation positions being enabled for the allocated pages. If it is determined that the perforation setting is not set, the CPU **201** proceeds to **S1403** to display the screen that the user tried to display in **S1401**, with a field related to the manual perforation setting not being grayed out. After that, the processing of the present flowchart ends.

[0107] By contrast, if it is determined that the perforation setting is set, the CPU **201** proceeds to **S1404** to display the screen that the user tried to display in **S1401**, with a field related to the manual perforation setting being grayed out. This means that, for example, the perforation position specification **404** in FIG. 4 is grayed out. It also means that, for example, while the perforation position specification **404** in FIG. 4 is not grayed out but selectable, the list **1202** in FIG. 12 is grayed out. In either case, the screen is displayed in such a manner that perforation settings cannot be manually configured. After that, the processing of the present flowchart ends.

[0108] The configurations described using FIGS. 13 and 14 helps prevent confusing a user who wants to configure perforation settings and therefore helps improve convenience.

[0109] FIGS. 15 and 16 to be described next show control performed in a case where conflicting

settings are configured and a setting configured later is prioritized.

[0110] FIG. **15** is a flowchart of screen control according to the present embodiment performed in a case where a user tries to finalize selection of the automatic perforation setting in addition to the setting for printing a plurality of images on a single sheet. The firmware **350** according to the processing of the embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**. Note that the present processing starts upon detection of an operation for configuring settings for execution of the print **311** or the copy **312**.

[0111] In **S1501**, the CPU **201** detects that a user tries to finalize selection of the automatic perforation setting in addition to the setting for printing a plurality of images on a single sheet. This is to detect, for example, pressing of the OK button **506** on the page consolidation function setting screen **501** with the page consolidation function being set and the checkbox **504** for automatically setting perforation positions being enabled for the allocated pages.

[0112] In **S1502**, the CPU **201** determines whether manual perforation settings have already been configured on a different screen. If it is determined that manual perforation settings have not been configured, the CPU **201** proceeds to **S1503** to finalize selection of the automatic perforation setting that the user tried to finalize in **S1501**. After that, the CPU **201** ends the processing of the present flowchart.

[0113] By contrast, if it is determined that manual perforation settings have been configured, the CPU **201** proceeds to **S1504** to cancel the perforation positions which have already been manually specified and configured and then finalizes selection of the automatic perforation setting that the user tried to finalize in **S1501**. After that, the CPU **201** ends the processing of the present flowchart.

[0114] FIG. **16** is a flowchart of screen control of the present embodiment performed in a case where a user tries to finalize selection of the manual perforation setting in addition to the setting for printing a plurality of images on a single sheet. The firmware **350** according to the processing of the present embodiment is stored in the HDD **204** of the image processing apparatus **103** and is loaded into the RAM **202** and executed by the CPU **201**.

[0115] In **S1601**, the CPU **201** detects that a user tries to finalize manual perforation settings. This may be, for example, a user trying to finalize manual perforation settings on the manual perforation position setting screen **1101** or the perforation settings list screen **1201**. In **S1602**, the CPU **201** determines whether the automatic perforation setting is already set on a different screen. If it is determined that the automatic perforation setting is not set, the CPU **201** proceeds to **S1603** to finalize the manual perforation settings that the user tried to finalize in **S1601**. After that, the CPU **201** ends the processing of the present flowchart.

[0116] Meanwhile, if it is determined that the automatic perforation setting is set, the CPU **201** proceeds to **S1604** to cancel the automatic perforation setting already set in addition to the setting for printing a plurality of images on a single sheet and then finalizes the manual perforation settings that the user tried to finalize in **S1601**. After that, the CPU **201** ends the processing of the present flowchart.

[0117] The above processing saves trouble in a case where, for example, a user who manually configured perforation settings changes their mind and changes the setting to the automatic perforation setting. Specifically, the above processing eliminates an operation for going back to the setting screen for configuring manual perforation settings (the manual perforation position setting screen **1101** or the perforation settings list screen **1201**). Conversely, by configuring manual perforation settings, the above processing saves the trouble of going back to the setting screen for the automatic perforation setting to cancel the automatic perforation setting already set. This improves convenience.

[0118] FIG. **17** is a diagram showing a setting change screen for changing, according to a user's preferences, control to be performed by the window management **309** according to the present

embodiment in a case of a conflict of perforation settings.

[0119] A setting change screen **1701** has a button **1702** which is a set value for prioritizing a setting set earlier and a button **1703** which is a set value for prioritizing a setting set later. The default set value on the setting change screen **1701** at the time of shipment is the set value of the button **1702**. In other words, at the time of shipment, the perforation setting set earlier is prioritized. The setting to prioritize an earlier setting or the setting to prioritize a later setting is finalized as a set value upon pressing of an OK button **1704** for finalizing the set value, with the corresponding button (the button **1702** or **1703**) being pressed. In a case where a Cancel button **1705** is pressed after either the button **1702** or **1703** is pressed on the setting change screen, the original set value set before a transition to this setting change screen is held.

[0120] Based on the setting configured on this setting change screen, the CPU **201** performs control shown in FIG. **13** or **14** in a case where the button **1702** is selected or performs control shown in FIG. **15** or **16** in a case where the button **1703** is selected.

[0121] As thus described, upon conflict between the manual and automatic perforation position settings, the present embodiment lets the user change which perforation setting to prioritize according to their preferences. Because different users may have different preferences regarding processing to be performed upon conflict, the present embodiment provides processing for avoiding a conflict according to a user's preferences by allowing the user to freely select processing to be performed upon conflict.

Embodiment 4

[0122] In the present embodiment, the image consolidation function and the perforation setting are set not on the main body of the multifunction peripheral, but on a PC using a printer driver or an application dedicated for printing, aiming to improve convenience in printing digital data on the PC, a cloud, or a server. The following describes the details.

<Screens Displayed According to Embodiment 4>

[0123] FIG. **18** is a software configuration diagram of the client PC **102** having software for a printer driver **1801** installed therein.

[0124] A driver information obtainment unit **1803** obtains and holds various kinds of information on the printer driver **1801**. The various kinds of information include information set on a user interface of the printer driver **1801**, like the one shown in FIG. **19**.

[0125] An apparatus information obtainment unit **1804** obtains apparatus information on the image processing apparatus **103** via the network. The apparatus information is, for example, information on the colors of toners in the image processing apparatus **103**.

[0126] A print data generation unit **1805** generates print data with which to instruct the image processing apparatus **103** to execute printing. Object generation for specifying rendering white regions necessary for printing in white is also performed by the print data generation unit **1805**.

[0127] The printer driver **1801** displays a printer driver screen **1901** on the panel of the client PC **102** as requested. Although described here as being incorporated in the client PC **102**, the printer driver **1801** may be configured to be incorporated in the server PC **101**.

[0128] FIG. **19** is a diagram showing a printer driver screen according to the present embodiment displayed by the printer driver **1801**, presenting setting items to be set for execution of print. The printer driver screen **1901** is displayed on the server PC **101** or the client PC **102**.

[0129] The printer driver screen **1901** in FIG. **19** includes a similar function to that on the page consolidation function setting screen **501** in FIG. **5**, and it is shown as a page layout setting **1902**. The page layout setting **1902** has a set value such as 2-up or 4-up, like the page consolidation function setting screen **501** does. Also, the printer driver **1801** has a checkbox **1903** for automatically setting perforation positions, like the page consolidation function setting screen **501** in FIG. **5** does. The checkbox **1903** for automatically setting perforation positions may be indented like in FIG. **19** so as to indicate that it is an optional setting for the page layout setting **1902**.

[0130] Processing performed in a case where the checkbox **1903** for automatically setting

perforation positions is checked is similar to the processing described earlier using FIGS. 6 and 7. In the event where the processing is performed, set values related to the printer driver are, along with other set values, transmitted from the server PC **101** or the client PC **102** to the image processing apparatus **103** as a PDL job.

[0131] As thus described, according to the present embodiment, the image consolidation function and the perforation setting can be configured not only on the main body of the multifunction peripheral, but also on a PC using a printer driver or an application dedicated for printing. Because the image consolidation function and the perforation setting are in-demand settings in printing digital data on the PC, a cloud, or a server, the above mode may improve convenience.

Other Embodiments

[0132] The setting screen described in Embodiment 3 using FIG. 17 is for a user to preliminary determine whether to prioritize a setting configured earlier or to prioritize a setting configured later between the automatic perforation setting and the manual perforation setting. However, the determination method is not limited to this. For example, in a case where either the automatic perforation setting or the manual perforation setting has already been configured earlier and then a user tries to configure the other setting, a notification screen may be displayed to notify that a different perforation setting has already been configured earlier. The user then may select which of the perforation settings to use on this notification screen.

[0133] Also, the automatic perforation setting and the manual perforation setting may be combined as needed. As one of such modes, for example, after the automatic perforation setting is configured, the automatically-set perforation lines can be adjusted manually, or a new perforation line can be added manually.

[0134] In the printing and processing system described above, printing or processing settings are configured on the operation unit **212** of the image processing apparatus or a setting screen displayed by a printer driver in the client PC or the like, and the image processing apparatus **103** executes printing and processing. However, the present disclosure is not limited to this. For example, printing may be executed by the image processing apparatus **103**, and processing may be executed by a processing apparatus. Also, a printing and processing apparatus may execute printing and processing in response to an instruction to print and process. In other words, each process is not limited to being executed by the one described in the embodiment.

[0135] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0136] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The

scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0137] This application claims the benefit of Japanese Patent Application No. 2024-031977, filed Mar. 4, 2024, which is hereby incorporated by reference wherein in its entirety.

Claims

1. An information processing apparatus comprising at least one memory and at least one processor which function as: a consolidation unit that configures a setting for consolidating a plurality of sets of image data on a single sheet and a determination unit that automatically determines a position to perforate the sheet based on the setting configured by the consolidation unit.
2. The information processing apparatus according to claim 1, wherein the consolidation unit is configured to configure the setting by receiving a first operation by a user, the determination unit is configured to perform processing for determining the position by receiving a second operation by the user, and the information processing apparatus further comprises a display control unit that displays a first setting screen as a setting screen for receiving not only the first operation but also the second operation.
3. The information processing apparatus according to claim 1, wherein the consolidation unit performs N-up or image repeat, the N-up being a function to allocate and print a plurality of originals on a single sheet, the image repeat being a function to allocate and print a particular image repeatedly on a single sheet in at least one of a vertical direction and a horizontal direction of the sheet.
4. The information processing apparatus according to claim 3, wherein in a case where the consolidation unit performs the N-up, the determination unit obtains a size of the sheet and a value of N (N is a multiple of 2) set for the N-up, the determination unit determines a position of a horizontal perforation line by repeating, $(N/2-1)$ times, setting a horizontal perforation line from a position corresponding to a length obtained by dividing a vertical size of the sheet by $N/2$, and the determination unit determines a position of a vertical perforation line by setting a vertical perforation line from a position corresponding to a length obtained by dividing a horizontal size of the sheet by 2.
5. The information processing apparatus according to claim 3, wherein in a case where the consolidation unit performs the image repeat, the determination unit obtains a size of the sheet, a value of X (X is a natural number) indicative of a vertical image repeat count, and a value of Y (Y is a natural number) indicative of a horizontal image repeat count, the determination unit determines a position of a horizontal perforation line by repeating, $(X-1)$ times, setting a horizontal perforation line from a position corresponding to a length obtained by dividing a vertical size of the sheet by X, and the determination unit determines a position of a vertical perforation line by repeating, $(Y-1)$ times, setting a vertical perforation line from a position corresponding to a length obtained by dividing a horizontal size of the sheet by Y.
6. The information processing apparatus according to claim 1, wherein in a case where a setting is configured to add margins to the sheet, the determination unit determines the position to perforate using vertical and horizontal sheet sizes excluding the margins.
7. The information processing apparatus according to claim 2, wherein the display control unit displays a second setting screen for receiving an operation for manually specifying a perforation position on the sheet.
8. The information processing apparatus according to claim 7, wherein in a case where the second operation is already received on the first setting screen, the display control unit performs control not to receive the operation for manually specifying a perforation position on the sheet on the second setting screen.
9. The information processing apparatus according to claim 8, wherein the control not to receive

the operation for specifying a perforation position on the sheet is to gray out, on the second setting screen, an entry field for configuring a perforation setting through a manual entry of a numerical value.

10. The information processing apparatus according to claim 9, wherein in a case where the second operation is already received on the first setting screen, the display control unit performs control to cancel the second operation received earlier upon receipt of the operation for manually specifying a perforation position on the sheet on the second setting screen.

11. The information processing apparatus according to claim 9, wherein in a case where the operation for manually specifying a perforation position on the sheet is already received on the second setting screen, the display control unit performs display control not to receive the second operation on the first setting screen.

12. The information processing apparatus according to claim 9, wherein in a case where the operation for manually specifying a perforation position on the sheet is already received on the second setting screen, upon receipt of the second operation on the first setting screen, the display control unit performs control to cancel the operation for manually specifying a perforation position received earlier.

13. The information processing apparatus according to claim 9, wherein the display control unit displays a third setting screen for a user to preliminarily determine whether to prioritize an operation received earlier or to prioritize an operation received later between an operation for implementing the determination unit performed on the first setting screen and the operation for manually specifying a perforation position performed on the second setting screen.

14. The information processing apparatus according to claim 13, wherein in a case where the operation received earlier is prioritized, the display control unit performs control so that the operation received later cannot be performed.

15. The information processing apparatus according to claim 13, wherein in a case where the operation received later is prioritized, the display control unit performs control to cancel the operation received earlier.

16. The information processing apparatus according to claim 1, further comprising an instruction unit that instructs a printing and processing apparatus to execute printing and processing based on the consolidation unit and the determination unit.

17. A method of controlling an information processing apparatus, comprising: configuring a setting for consolidating a plurality of sets of image data on a single sheet and automatically determining a position to perforate the sheet based on the setting configured in the configuring.

18. A non-transitory computer readable storage medium storing a program which functions in an information processing apparatus and causes the information processing apparatus to function as: a consolidation unit that configures a setting for consolidating a plurality of sets of image data on a single sheet and a determination unit that automatically determines a position to perforate the sheet based on the setting configured by the consolidation unit.
